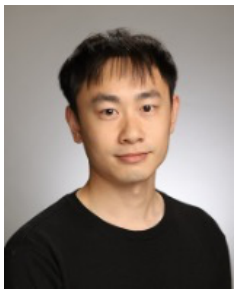
Dr. Guangping Zeng

Postdoctoral Researcher
Deep Sea Drilling Center, Guangzhou Marine Geological Survey
South China Sea Institute of Oceanology, Chinese Academy of Sciences
Email: zengguangping22@mails.ucas.ac.cn

Research Interests



My research examines Earth material cycles across environments, resources, oceans, crust, and mantle. I combine isotope geochemistry, petrology, sedimentary records, ocean drilling, and data synthesis to trace how carbon, water, volatiles, and nutrients are produced, transported, transformed, and recycled. Current projects connect Makran sediment recycling, Pakistan magmatism, South Poyang Basin paleoenvironmental and resource records, and South China Sea spreading-ridge basalt geochemistry.

Academic Appointments & Training

Postdoctoral Researcher (Advisor: Prof. Zhen Sun) 2025-present
Deep Sea Drilling Center, Guangzhou Marine Geological Survey (host institution)
South China Sea Institute of Oceanology, Chinese Academy of Sciences (affiliation)

Visiting Scholar (Advisor: Prof. Susanne M. Straub) Apr. 2024-Apr. 2025
Lamont-Doherty Earth Observatory, Columbia University

Ph.D., Marine Geology (Advisor: Prof. Yunying Zhang) 2022-2025
South China Sea Institute of Oceanology, Chinese Academy of Sciences

M.S., Sedimentology and Paleogeography (Advisor: Dr. Jun Wang) 2019-2022
Institute of Sedimentary Geology, Chengdu University of Technology

Research Experience

South China Sea Spreading-Ridge Basalt Geochemistry 2025-present
Developing an exploratory project using fresh basalt samples from a South China Sea spreading ridge, combining radiogenic and stable isotope tracers to investigate mantle sources, oceanic crust formation, and deep material cycling.

Makran Arc: Sediment and Carbon Recycling 2024-present
Integrating petrology, isotope geochemistry, and thermodynamic modeling to constrain sediment recycling, carbon return, and volatile fluxes in the Makran system.

Cenozoic Volcanism in Pakistan: Petrogenesis and Tectonic Evolution 2022-2025
Applying geochronology, petrology, and geochemistry to investigate magma genesis and its tectonic implications from subduction to continental collision.

South Poyang Basin: Paleoenvironmental Records 2019-2022
Combining sequence stratigraphy and geochemical analyses to reconstruct paleoenvironmental evolution and constrain controls on organic matter enrichment.

Lab Skills

Geochemistry & Microanalysis: SIMS, MC-ICP-MS, LA-ICP-MS, SEM, XRF
Data Analysis: Python, MATLAB, ArcGIS

Honors and Awards

AGU Student Travel Grant $1,000 (2024)
UCAS Joint Ph.D. Training Fellowship $24,000 (2024-2025)
National Encouragement Scholarship (2020-2022)
Outstanding Graduation Thesis Award (2019)
Excellent Camper, Nanjing Institute of Paleontology, CAS (2018)
University People's Scholarship Award (2017)
University Student Summer Social Practice Group Award (2016)

Publications

Peer-reviewed Articles

Gardezi, S.A.H., Luan, X., Li, F., Lin, J., Zeng, C., Zeng, G., Zhang, Y., Yang, X., Khan, M.J., Sun, Z., 2026. Structural evolution of mud-intrusive and mud-extrusive system in the Makran accretionary wedge: Processes and patterns. Marine Geoscience and Energy Resources 193, 207833. https://doi.org/10.1016/j.marger.2026.207833

Zeng, G., Zhang, Y., Sun, Z., Mehmood, M., Gardezi, S.A.H., Miao, X., Sun, L., Li, X., 2026. Unraveling the India-Eurasia Collision Pattern: Insights from Subduction to Collision-Related Magmatism in the Chagai Hills. J. Earth Sci. 37, 1055-1069. https://doi.org/10.1007/s12583-025-0244-z

Zeng, G., Wang, J., Zhang, C.G., Zhao, Q., Wu, S., 2022. Paleogeographic environment characteristics of the Guanshan Member of the Leping Formation in the Erjiacun Depression, southern Poyang Basin. Fault-Block Oil & Gas Field 29(2), 199-206 (in Chinese).

Conference Abstracts

Zeng, G., Zhang, Y., Straub, S.M., et al., 2024. Sediment recycling at the Makran arc: Insights from whole-rock elemental and Sr-Nd-Pb isotope systematics. Abstract, AGU Fall Meeting 2024.

Manuscripts under Review

Zeng, G., Zhang, Y., Sun, Z., et al. Efficient return of subducted organic carbon in the Makran arc volcano. Under review.